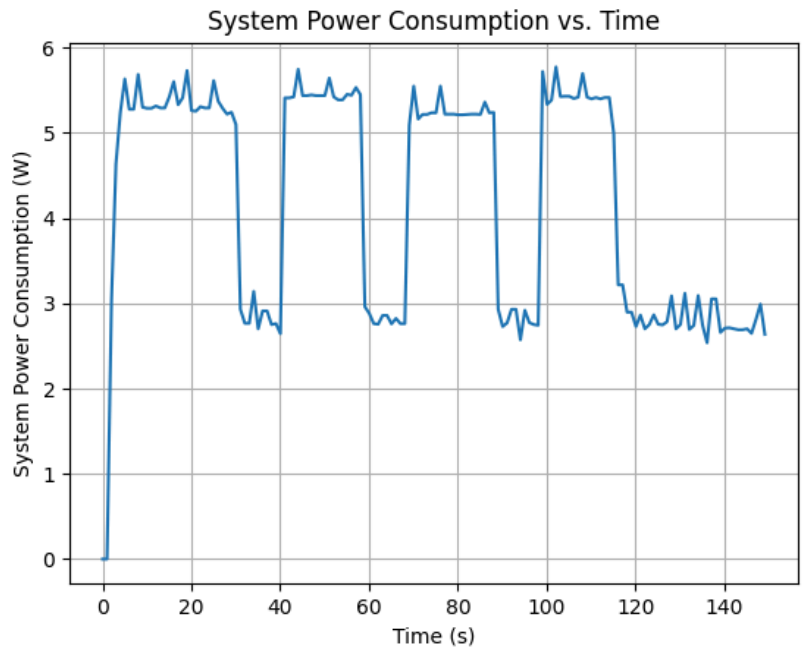
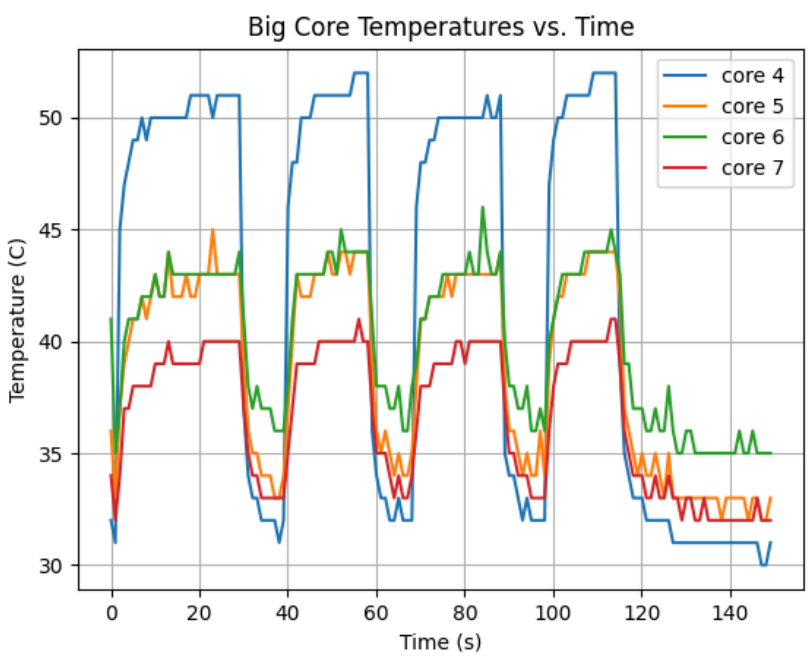
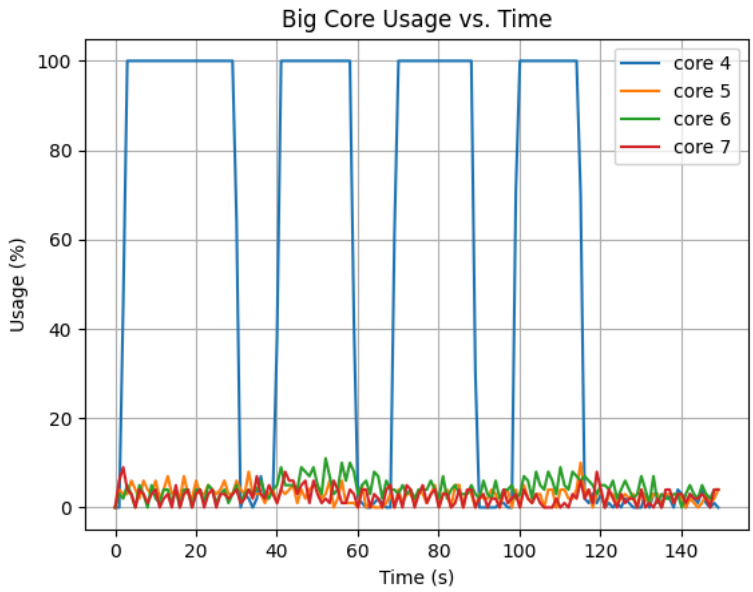
Suraj Swamy
Laksh Gulati

Problem 1

Question 1







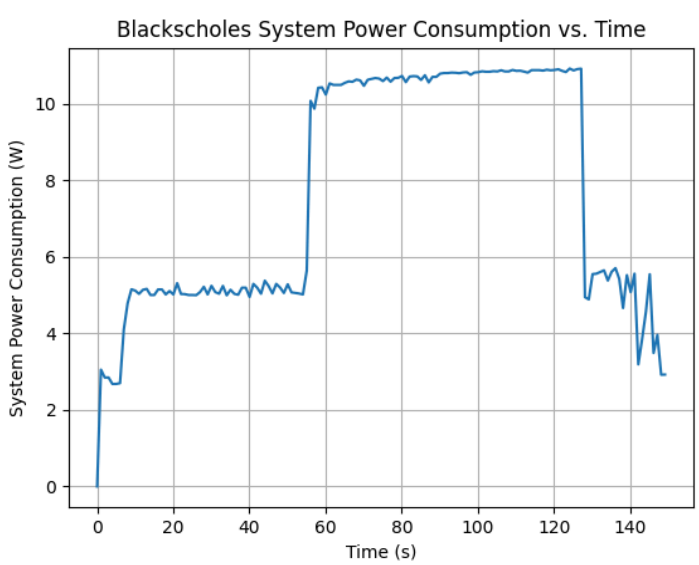
Question 2

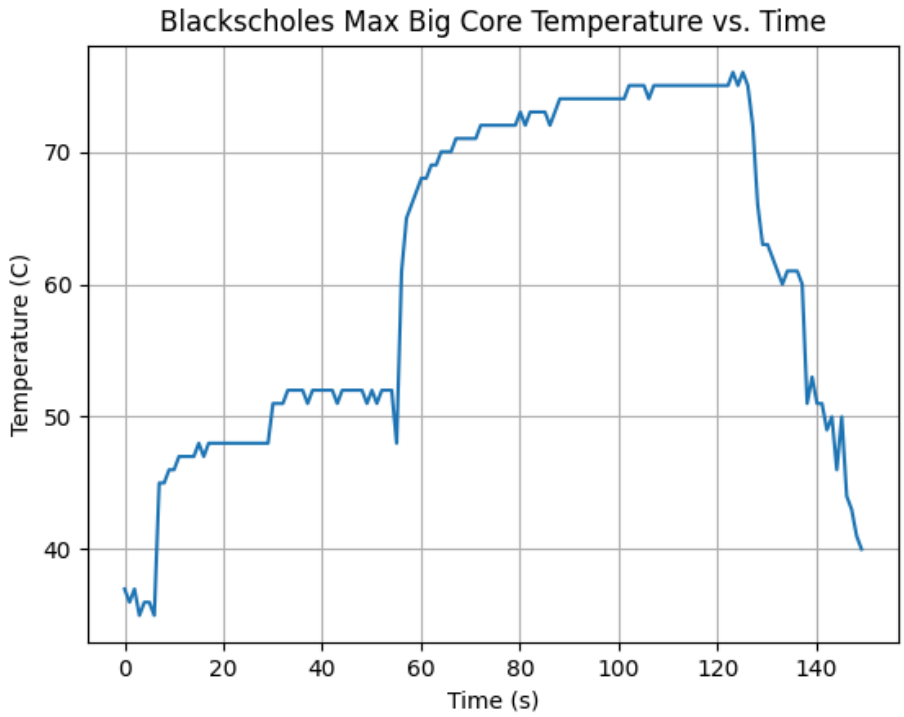
4 phases of benchmark execution

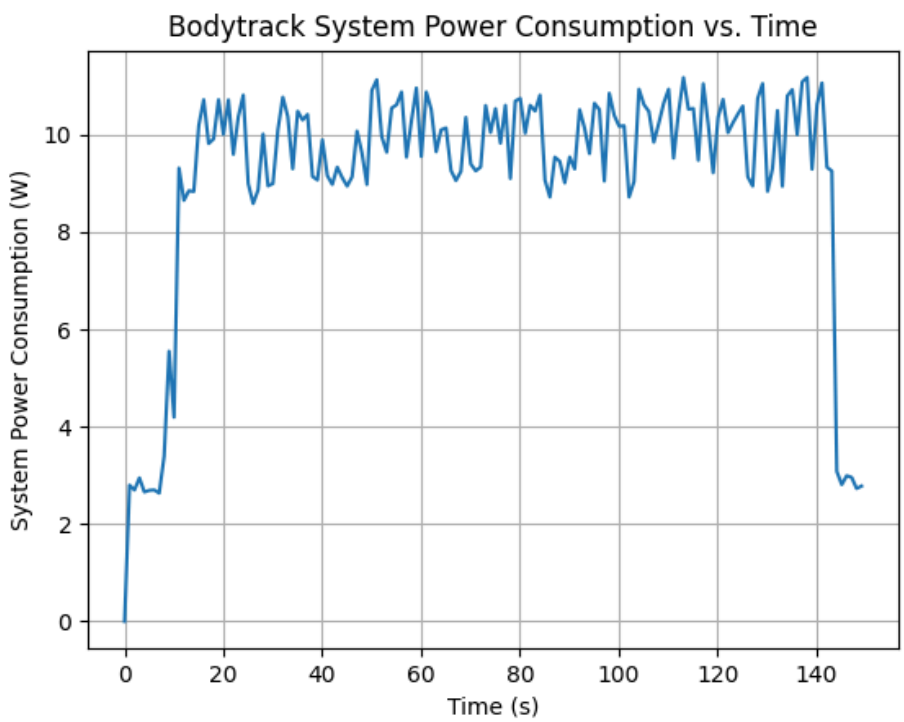
Question 3

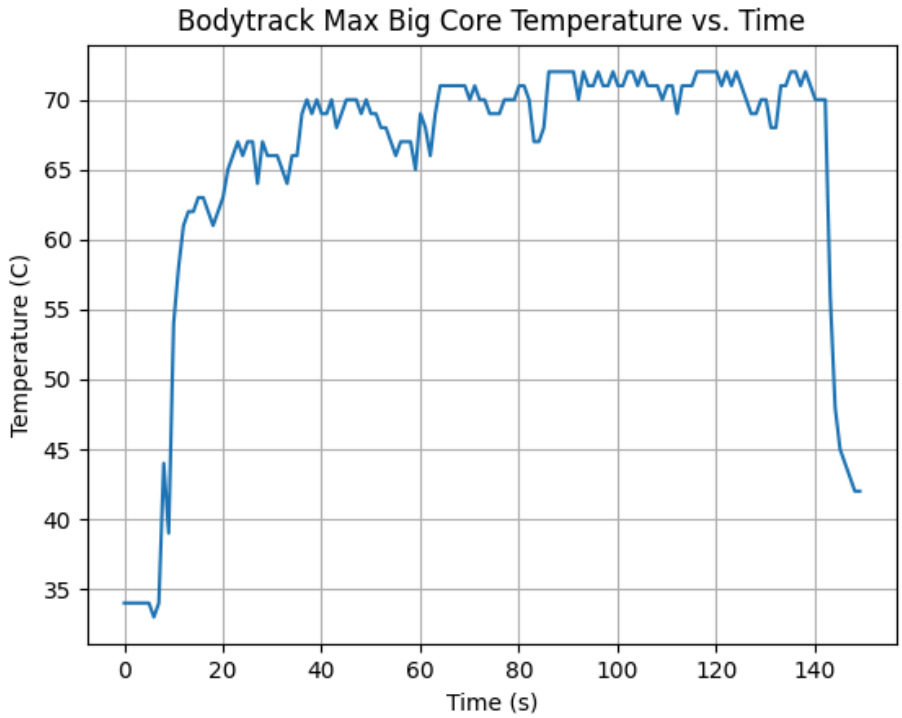
Table 1

Benchmark	Run time (s)	Avg. power (W)	Avg. max temp (C)	Max temp (C)	Energy (J)
blacksholes	133.980	7.624	60.73	76.0	1021.43
bodytrack	133.177	9.164	65.58	72.0	1220.43




Blackscholes Max Big Core Temperature vs. Time
Temperature (C)
Time (s)
70
60
50
40
0
20
40
60
80
100
120
140


Bodytrack System Power Consumption vs. Time
System Power Consumption (W)
Time (s)
10
8
6
4
2
0
0
20
40
60
80
100
120
140



Problem 2

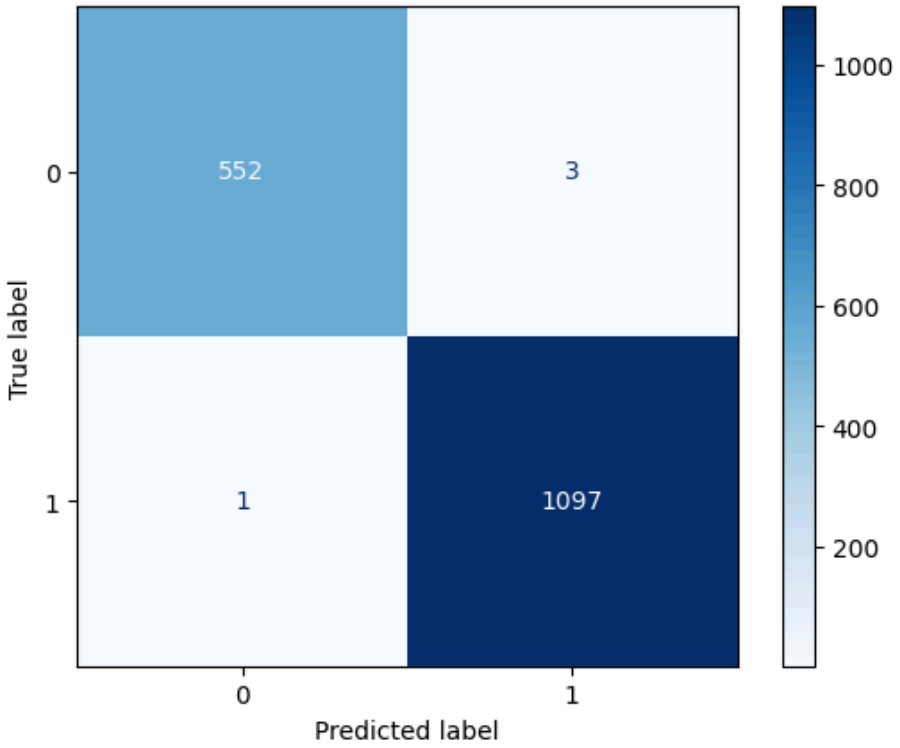
Question 1

Table 2

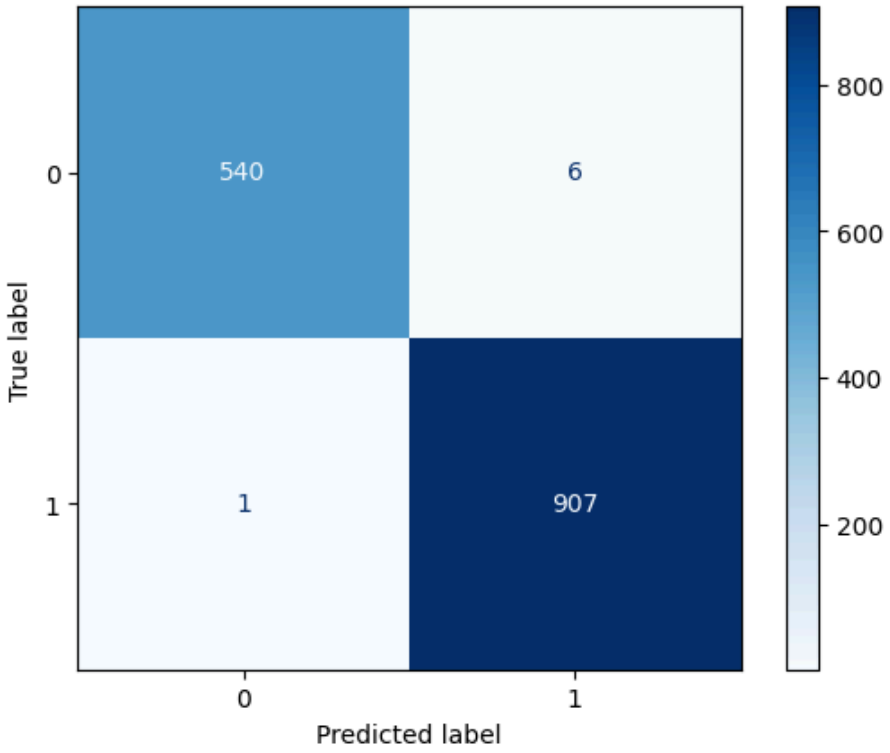
Benchmarks	Accuracy (%)	Avg. Precision	Avg. Recall	Avg. F-1 Score
blackscholes	0.9976	1	1	1
bodytrack	0.9951	1	1	1

The classifier is surprisingly very accurate on the data. There are almost no false positives or false negatives which is why the precision and recall are 1.

Blackscholes Confusion Matrix



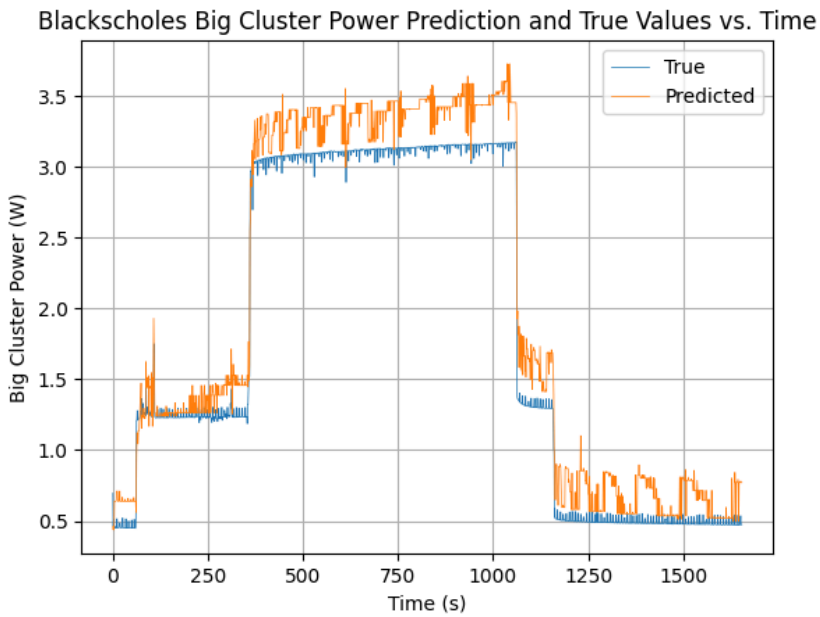
Bodytrack Confusion Matrix

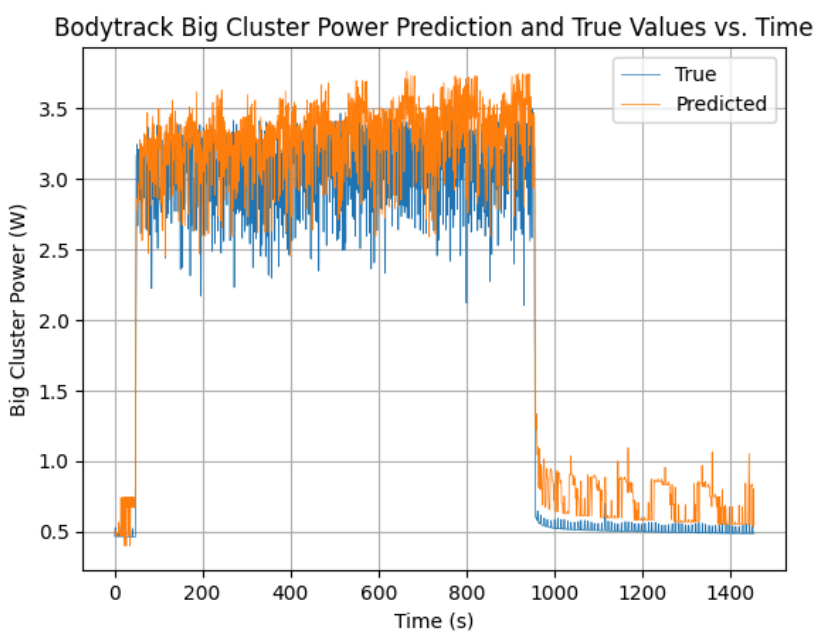


Question 2

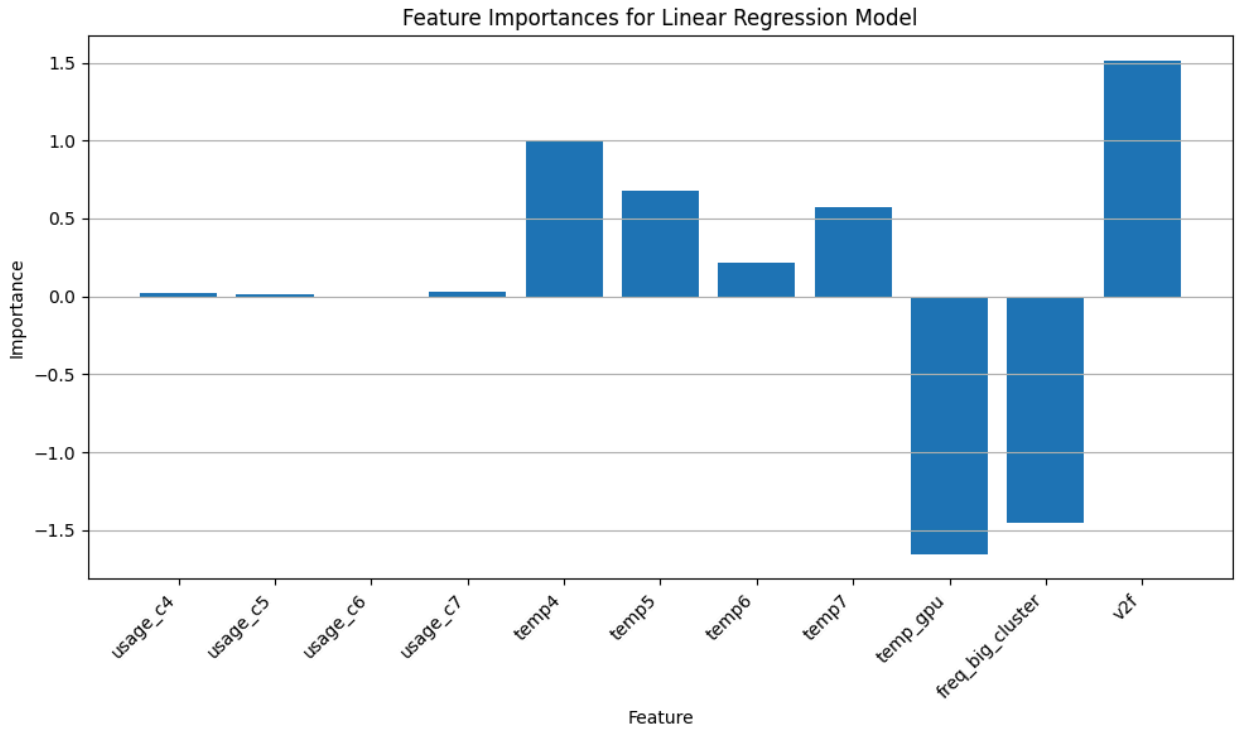
Table 3

Dataset	training	blackscholes	bodytrack
R^2	0.98698	0.95663	0.92243
MSE	0.01015	0.05942	0.12156





Question 3



Feature: usage_c4, Score: 0.02348
Feature: usage_c5, Score: 0.01572
Feature: usage_c6, Score: -0.00097
Feature: usage_c7, Score: 0.02974
Feature: temp4, Score: 1.00116
Feature: temp5, Score: 0.68157
Feature: temp6, Score: 0.21325
Feature: temp7, Score: 0.57254
Feature: temp_gpu, Score: -1.65602
Feature: freq_big_cluster, Score: -1.45737
Feature: v2f, Score: 1.51417

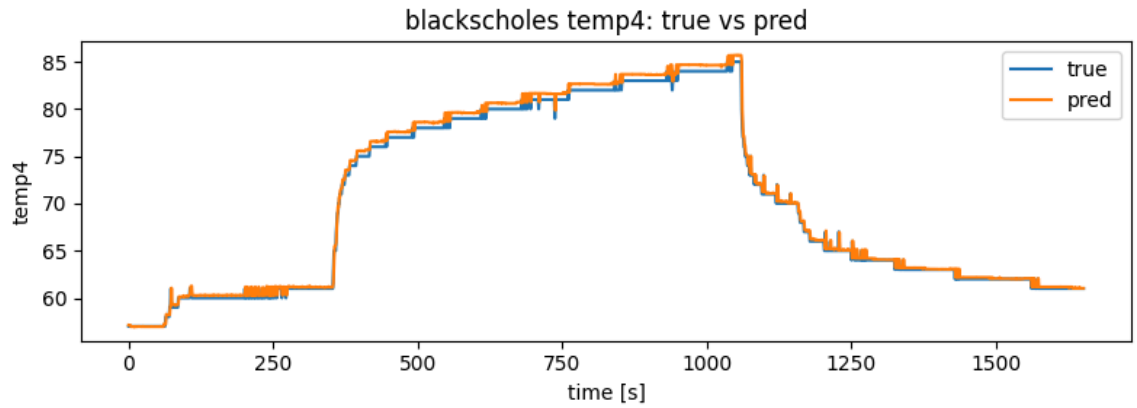
The top three positive features are temperature of core 4, temperature of core 5, and V_{dd}^2f. I observe that core temperatures are valuable features to the model. This is likely because core temperatures rise with more computational activity, so high core temperatures are indicative of greater power consumption and vice versa. I also notice that V_{dd}^2f is an important feature to the model, likely because power consumption is strongly related to input voltage. Lastly, I notice that gpu temperature is a significant negative feature to the model. This is likely due to workload distribution between the CPU and the GPU. If the GPU does more computation, and its temperature rises accordingly, then the CPU is likely doing less computation.

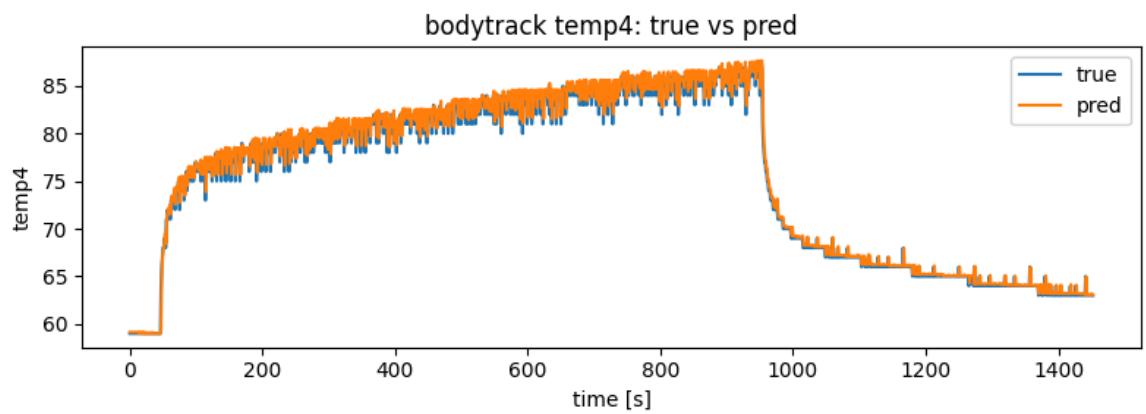
Problem 3

Question 1

Table 5

Dataset	Test MSE (Core 4)	Test MSE (Core 5)	Test MSE (Core 6)	Test MSE (Core 7)
blacksholes	0.1808	0.1168	0.2542	0.0694
bodytrack	0.3106	1.4564	0.2587	0.1484





Question 2

Two ways we can improve the performance of the model are:

1. Hyperparameter tuning- right now we are using the given values from the documentation, and we can try to find the optimal values
2. Time series model- if we use a model with built in temporal layers (ex. RNN/LSTM), we may be able to improve performance.